Highlights

The numerical certificate behind the two-thirds theorem, checked without Lean, and the file it still needs*

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- The formalisation's one numerical certificate holds over its printed integers by a route that shares nothing with Lean; its hypothesis EnclOK rests on an unpublished file, which we ask for.
- The Montgomery–Taylor and ξ' constants are decided by exact rational arithmetic, no interval kernel; the ξ' rationals re-derive from their definitions.
- A truncated print is not a rounded one: the abstract's 0.8362 is true as a lower bound and false as a rounding.
- Of 65 sentences the story's sources print, 10 reduce to decidable claims; every theorem is refused, with the rule.

The numerical certificate behind the two-thirds theorem, checked without Lean, and the file it still needs

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Abstract

In 2026 a proof that more than two thirds of the nontrivial zeros of the Riemann zeta function are simple and on the critical line was found by a language model, verified by Alpöge and Furman, and formalised in Lean. The formalisation has one numerical input: 256 integer enclosures of a form factor at scale 2^{140}, kernel-checked downstream of a hypothesis, EnclOK, that the kernel does not check and that rests on a law file the release describes as available from the authors. We re-check the certificate from its integers with two programs that share no code with Lean, state exactly what that establishes and what it does not, and ask for the file. On the way, the Montgomery–Taylor constant and the four ξ' constants of the paper are decided by exact rational arithmetic alone, with no interval kernel; the paper's abstract prints one constant truncated where a rounding would print another digit, true as a lower bound; and a survey of everything the story's sources print—the manuscripts, the Lean files, the research post, a video's captions—finds that 10 of 65 sentences are finite claims at all. No theorem is touched. Every number here is read from a public record at build time.

Keywords: Riemann zeta function, simple zeros, Lean, formal verification, independent verification, exact rational arithmetic, machine-generated mathematics

1. The certificate, and what it rests on

Alpöge and Furman (Alpöge and Furman, 2026) prove that at least two thirds of the nontrivial zeros of ζ, counted with multiplicity, are simple

*Draft v0.2 of 2026-10-22, a companion to the cert-machine method paper, built from repository commit 61aae8eb42f1. Every number and count is read at build time from the records named in the Data availability statement; a record that no longer supports a sentence stops the build. Nothing here is a theorem, and nothing here bears on the truth of the theorems the story is about.

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and on the critical line, and that at least five sixths are distinct; with the Montgomery–Taylor window (Montgomery, 1973; Carneiro et al., 2017) the constants become $2 - c_{\mathrm{MT}}^{-1} = 0.67250\ldots$ and $\frac{1}{2}(3 - c_{\mathrm{MT}}^{-1}) = 0.83625\ldots$. The argument was discovered and written by Claude, an AI developed by Anthropic; the authors verified it and take responsibility for it, as the paper's own dedication says. Anthropic formalised the result in Lean (Anthropic, 2026b) and described the work in a research post (Anthropic, 2026a); a video explained the checking to a wide audience (Adem B10, 2026). The formalisation's audit prints the axioms of every main statement; they are Lean's three standard ones.

One part of the formalisation is numerical. The bandwidth-one ceiling, `Zeta23/PairCeiling/`, is the stability inequality behind the paper's remark on the optimality of its method: a bound on the simple-point fraction of any law of 256-periodic marked configurations whose form factor is near the CUE values on the band. Its input is the generated file `Zeta23/PairCeiling/LawN256.lean` (26,084 bytes, sha256 `11ae9f7dcc17`), whose header states that it was computed by 70-digit interval arithmetic at scale $K = 2^{140}$ and holds enclosures $\mathrm{lo}_j \le K \cdot S(j) \le \mathrm{hi}_j$, $j = 1, \ldots, 256$, of the grid form factor of a law recorded in a certificate file, `cert_N256_blk_b128m.json` (sha256 `cc3de9917db4`), "available from the authors". The enclosures are 43-digit integers, every one exactly 1 unit wide. The row certificate they satisfy is

$$|256\,S(j) - j| \le \tau := 3{\cdot}10^{-40} \quad (0 < j < 256), \qquad |D(1)| \le 0.82395317, \tag{1}$$

with the closed-band row $j = 256$ free; the law's simple-point fraction is the exact rational $p_0 = 1090925899942130358809523019581605440819$ $7/160000000000000000000000000000000000000000$, that is $0.68182868746\ldots$, which the paper prints as $p_0 \le 0.6818287$.

The Lean side is as the audit prints it. `LawN256_check`, the kernel check of (1) over the enclosures, depends on the axiom `propext` alone. `ceiling_law256` and the other ceiling theorems depend on the three standard axioms and, as the audit says of all of them except the two kernel checks, carry the displayed hypothesis `EnclOK`: that the true form factor of the law lies in the enclosures. That hypothesis is about the law itself, the weights and positions in `cert_N256_blk_b128m.json`, and the release does not publish that file. The audit is exemplary in saying so. But it means that the one place where the formalisation's numerical certificate meets the real numbers is checked by interval arithmetic outside Lean, from bytes no one outside can read.

2. What was checked without Lean

We decoded the integer literals of `LawN256.lean` into a JSON file beside it, naming the Lean file's digest, and decided three claims over those integers

with two programs, one in Python and one in JavaScript, that share no code with each other or with Lean and evaluate in exact rational arithmetic (Table 1; the programs are described in Section 5):

- the band rows, $|256\,S(j) - j| \le \tau$ over the 255 rows, with both ends of each enclosure tested: the worst row sits at $1.836 \cdot 10^{-40}$ from its target against the tolerance $3 \cdot 10^{-40}$;
- the edge bound, $|D(1)|$ with $D(1) = \sum_{j \le 256} S(j)/256 - \frac{1}{2}$ evaluated at both ends of the enclosures, which is exact because $D(1)$ is monotone in each $S(j)$: $|D(1)|$ at the ends is $0.82395316071\ldots$, under the printed bound by $9.28 \cdot 10^{-9}$;
- p_0, the file's rational against the printed bar.

All three hold, in milliseconds. Beside them, three pieces of the paper's own arithmetic on the ceiling were decided from the file's p_0 and τ: $p_0 - \frac{2}{3} = 0.0151620207\ldots$ is within the paper's 0.016; the decimal constant the Lean prints beside the ceiling, $1/(6 \cdot 256^2) + \tau/512 = 2.5431315104 \cdot 10^{-6}$, is under the paper's rounded $2.55 \cdot 10^{-6}$; and the paper's bound for any window with $|r'(1)| + \int |r''| < 8.2$, $p_0 + 2.55 \cdot 10^{-6} \cdot 8.2 = 0.68184959746\ldots$, is under its printed 0.6819.

claim, as printed	source	decided by	word	s
$\lvert 256 \cdot S(j) - j\rvert \le 3 \cdot 10^{-40}$ $(0 < j < 256)$	zeta23/ README.md	the 255 band rows, each end against the tolerance	CERTIFIED	0.014
$\lvert D(1)\rvert \le 0.82395317$	zeta23/ PairCeiling/ CeilingLaw256.lean	$\lvert D(1)\rvert$ at both ends of the enclosures	CERTIFIED	0.003
$p_0 \le 0.6818287$	arXiv/paper-v5-draft18.tex	the exact rational the file prints against the bar	CERTIFIED	0.001
$c_1^* = 0.7532960\ldots$	zeta23/ ChallengeDeps.lean	the series certificate, $K = 8$	CERTIFIED	0.003
$2 - 1/c_1^* = 0.67250\ldots$	zeta23/ formalization.yaml	the series certificate	CERTIFIED	0.002
$(3 - 1/c_1^*)/2 = 0.83625\ldots$	zeta23/ formalization.yaml	the series certificate	CERTIFIED	0.002
$2 - c_{\mathrm{MT}}^{-1} = 0.67250\ldots$	arXiv/paper-v5-draft18.tex	kept on the closed form: transcendental head, admitted, not decided	undecided	0.0
EnclOK	zeta23/ PairCeiling/ LawN256.lean	no bytes, no claim	skipped	—

Table 1: The certificate run: 8 claims, 6 decided and holding (CERTIFIED), 1 admitted and undecided (the paper's own closed form, kept on that route on purpose; see Section 3), 1 with no bytes to decide from.

A property of the file. Every band enclosure contains $j \cdot K/256$ exactly: 124 rows at the upper end, 131 at the lower, counted from the decoded integers and decided as $|2\,j\,K/256 - \mathrm{lo}_j - \mathrm{hi}_j| \le 1$ over the 255 rows. The law's form factor is thus consistent with $S(j) = j/256$ on every band frequency to 2^{-140}: the enclosures do not merely lie within $3 \cdot 10^{-40}$ of the CUE values, they sit on them, and $D(1) = S(256)/256 - 1/512$ to the same precision, with $256\,S(256) = 54126.5943404763\ldots$. This is most likely so by construction, an optimal law of a linear programme whose constraints are the CUE band values being expected to meet them exactly, and it is reported as a property of the file, not as a finding about the method.

What this establishes, and what it does not. The three checks are decided from the same integers the Lean kernel accepted; the independence is in the checking code, not in the data. They establish that the row certificate (1) holds over the printed enclosures by a route that does not pass through Lean, its elaborator or its `decide` tactic, a no-shared-code concordance with `LawN256_check`. They do not establish that the enclosures enclose anything. That is EnclOK, and it waits on the law file: there is no reading of any published byte that decides it, because the bytes it is about are not there.

The request. A request for `cert_N256_blk_b128m.json`, with its interval-arithmetic script if the authors are willing, has been drafted and will be posted on the formalisation's repository. With the file, the enclosures can be re-derived by an independent interval arithmetic from the law's weights and positions, and the certificate becomes checkable end to end by exactly the pipeline described here with one more claim. Until then it is a conditional certificate, for the kernel and for us alike.

3. Two families of constants, decided by rational arithmetic

The Montgomery–Taylor constant. The formalisation defines $c_1^* = \sqrt{2}\tan\vartheta/(1 + \vartheta\tan\vartheta)$ at $\vartheta = 1/\sqrt{2}$, the reciprocal of the paper's $c_{\mathrm{MT}}^{-1} = \frac{1}{2} + \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}}$. An exact checker over rationals cannot evaluate a tangent, and an interval kernel would be the usual answer. None is needed. With $\vartheta^2 = \frac{1}{2}$ the Maclaurin series give $\sin\vartheta = \vartheta s$ and $\cos\vartheta = c$ for

$$s = 1 + \sum_{k\ge1} \frac{(-1)^k}{2^k(2k+1)!}, \qquad c = 1 + \sum_{k\ge1} \frac{(-1)^k}{2^k(2k)!}, \tag{2}$$

alternating series of rationals whose terms fall by a factor of at least twelve at every step; hence $\tan\vartheta = \vartheta s/c$ and, since $\sqrt{2}\,\vartheta = 1$,

$$c_1^* = \frac{2s}{2c+s}, \qquad 2 - \frac{1}{c_1^*} = \frac{3}{2} - \frac{c}{s}, \qquad \frac{1}{2}\Big(3 - \frac{1}{c_1^*}\Big) = \frac{5}{4} - \frac{c}{2s}. \tag{3}$$

For an even truncation order K the partial sums bracket the limits, $s_{K+1} \le s \le s_K$ and $c_{K+1} \le c \le c_K$, and $2s/(2c+s)$ increases in s and decreases in c on positive values, so $2s_{K+1}/(2c_K + s_{K+1}) \le c_1^* \le 2s_K/(2c_{K+1} + s_K)$. At $K = 8$ this is the enclosure [0.75329606785607067702, 0.75329606785607067722], of width $1.96 \cdot 10^{-19}$, in sums, products and powers over integer ranges. The three prints of the sources, $c_1^* = 0.7532960\ldots$, $2 - 1/c_1^* = 0.67250\ldots$ and $(3 - 1/c_1^*)/2 = 0.83625\ldots$, hold as truncation boxes, and so do the two bounds one unit of the eighth digit either side, $c_1^* \le 0.75329607$ and $c_1^* \ge 0.75329606$. This is elementary; the point is only that it removes the kernel, and that the paper's own print of $2 - c_{\mathrm{MT}}^{-1}$, kept on the closed-form route on purpose, stands admitted and undecided in Table 1 to show what the checker cannot do without it.

The ξ' constants. The paper's Remark 7.1 states that at least 0.85838 of the zeros of ξ' are simple and on the critical line and at least 0.92919 are distinct (flat window), 0.86864 and 0.93432 with the quartic window $v(s) = 1 - \frac{7}{100}(2s)^2 - \frac{51}{200}(2s)^4$; the formalisation proves eight-digit versions, 0.85838371, 0.92919185, 0.86864017, 0.93432008, from a certificate every number of which is rational: the diagonal density $D_1(s) = s - 4s^2 + \sum_{k\ge 0} a_k s^{2k+3}$ with $a_k = 2 \cdot 4^{k+1} k!/(2k+2)!$, truncated at nine terms ($k = 0, \ldots, 8$) with the certified tail $\varepsilon_9 = a_9 \cdot 231/211 = 3.424 \cdot 10^{-7}$ (the coefficient ratio is at most $20/231$ from $k = 9$ on); the window integral $J(v) = 2\int_0^1 D_1^{(9)}(r)\,(v{*}v)(r)\,dr$ with $v{*}v$ the window's autocorrelation; $\kappa_9(v) = (\int v^2 + J(v))/(\int v)^2$; and the proportions $2 - \kappa_9 - \varepsilon_9$ and $\frac{3}{2} - (\kappa_9 + \varepsilon_9)/2$. We re-derived every number from those definitions in exact fractions, the quartic window's degree-nine autocorrelation polynomial included, twice and independently (once in the translator's session, once in the generator of this paper's numbers, which stops the build if any differs), and each equals the Lean's computer-algebra-generated closed form:

$$J_{\mathrm{flat}} = 12517210259/88388425125, \qquad \varepsilon_9 = 1024/2990212875,$$
$$\kappa_9^{\mathrm{flat}} = 100905635384/88388425125,$$
$$J_{\mathrm{quartic}} = 29965280330934234270211/28599109093767712500000 0,$$
$$\kappa_9^{\mathrm{quartic}} = 27724414054746915416833 6/24505397663619131972212 5.$$

The four eight-digit constants hold with margins $3.59 \cdot 10^{-9}$, $6.79 \cdot 10^{-9}$, $3.59 \cdot 10^{-9}$ and $6.79 \cdot 10^{-9}$; the paper's five-digit prints hold; $P_{\mathrm{flat}} = 16008819950038/18649957701375 = 0.8583837135\ldots$ exactly, and a control one unit of the eighth digit above it, $P_{\mathrm{flat}} \ge 0.85838372$, is refuted as it must be. (The identity claims cite the session's derivation note as their source; the Lean files of the ξ' certificate are being pinned by the producer, and the record will name them when they are.)

Two corrections the machine made. Both re-derivations began wrong, and both times the exact decision was the correction. The grammar's first reading gave a print cut off with an ellipsis, $0.7532960\ldots$, the half-unit rounding box; the checker then refuted the true print and certified a false $0.7532961\ldots$, and it was right, since $c_1^* = 0.75329606785\ldots$ lies in [0.7532960, 0.7532961) and not within half a unit of 0.7532961. An ellipsis is a truncation, and the grammar now reads it so. And a first pass took "`D1trunc 9`" as ten terms and reproduced none of the printed rationals; nine terms reproduce all of them, and the generator checks that a ten-term reading still matches none. We report both because they are the kind of error an exact decision catches and a careful reader does not.

4. A print, and a rule

The paper's abstract prints: "With the Montgomery–Taylor window the constants become 0.6725 and 0.8362." Its theorem prints $0.67250\ldots$ and $0.83625\ldots$. The exact second constant is $0.836250351839705822\ldots$, which lies above the box the four printed digits allow as a rounding, [0.83615, 0.83625]: 0.8362 is a truncation, and a rounding would print 0.8363. As a lower bound on a proportion the truncated print is still true, a weaker true statement, and the paper says nothing else about it. The rule it illustrates is general: a print cut off with an ellipsis and a print rounded to four digits are different claims, and a reader with exact arithmetic will tell them apart. In our grammar the two are read differently, and Table A.7 in the appendix shows the claim "$(3 - 1/c_1^*)/2$, published as 0.8362" refuted under the rounding reading and its companion 0.6725 holding.

5. How the claims were read, and how much of the story is finite

The claims above were not typed by hand. They were extracted from the sources by a translator with a closed grammar, `claimbox` (Toledo, 2026b), and decided by an exact checker, cert-machine, whose grammar of three verdicts with witnesses is described in the method paper this is a companion to (Toledo, 2026a, the method paper). The translator is a standard-library pattern matcher, with no model and no parser framework. A sentence is read only if it matches one of a short list of shapes: a trailing finite range; |EXPR| REL NUMBER; EXPR = NUMBER…, a value printed to its digits and cut off; NAME REL NUMBER; a constant "published as" a number; an exact identity NAME = RATIONAL; a few others for other corpora. The number is kept as the source spells it and read as the exact rational it means. Everything else is refused with a rule: an asymptotic, an unbounded quantifier, an approximation sign, two relations in one sentence, no printed quantity, prose. The object the translator emits carries the line of the source that prints the claim with the file's digest, the pin of the bytes

and a sentence saying how they were decoded, the predicate, the hypotheses under which the predicate is the claim, and the shape a witness against it would take; the checker resolves the pin, reads the data as stated, evaluates the predicate over exact rationals and prints the word with a witness. Two checkers, in Python and in JavaScript, share no code and agree on every object here. Neither runs the claimant's code, reads a proof, or evaluates a transcendental function: a closed form over exp, log, sin or cos is admitted and left undecided.

Put to the whole story, the grammar is a measuring instrument. The 65 sentences the sources print were extracted byte for byte, tagged by author, and read: 10 match a shape and 55 are refused (Table 2), 43 as prose with no claim shape, 6 with no printed quantity, 3 as ambiguous, 2 as asymptotic, 1 as an approximation. Every theorem is among the refused: a statement about the limit inferior of a ratio of counting functions is not a finite claim, and the grammar says so rather than inventing one. The 10 read are the certificate's inequalities and the digits prints of Sections 2 and 3, and nothing else. The 11/12 companion manuscript of the `openai/math` release (OpenAI, 2026) (784,503 bytes, sha256 `f919b57829b1`; 3,073 lines by `pdftotext` 26.07.0) states 34 numbered theorems, lemmas, propositions and corollaries and prints no decimal constant at all; it is a proof for human readers, and nothing in it is a claim this pipeline can touch. The same is true of eighty phrasings of 8 of the story's claims written by a fresh AI agent in the voices of a paper, an abstract, a README, a Lean docstring, a blog, a tweet, a referee report and a chat assistant: 3 read, 77 refused, because a claim is read at the head of its sentence or not at all, and nothing in the grammar was extended for a phrasing no pinned source prints.

rule	printed sentences	AI phrasings
prose: no claim shape	43	66
no printed quantity	6	1
ambiguous: two readings	3	3
asymptotic ($o(1)$, as $T \to \infty$)	2	0
an approximation sign	1	4
an unbounded quantifier	0	3

Table 2: The refusals by rule, over the 55 refused printed sentences and the 77 refused AI phrasings. The 10 sentences read are listed in Table A.4.

That the finite layer of a celebrated result is this thin is, we think, the most useful number in this paper. It is not a criticism of the result; a proof is not a list of inequalities. It is a measurement of what independent machine checking can reach in a story of this kind without reading proofs, and the answer, here, is one certificate and a handful of printed constants. The battery that holds the instrument to account, planted claims with their expected words, the author's own claims as a mathematician with expectations

recorded before each run, and 782 planted defects at scale on the certificate with 511 controls undisturbed, is in Appendix A.

6. Conclusion

The formalisation's numerical certificate holds over its printed integers by a route that shares nothing with Lean; its hypothesis EnclOK rests on a file that only its authors can read, and we have asked for it. The constants that looked as if they needed an interval kernel needed only an alternating series and the certificate's own definitions. One print in the abstract is a truncation. Of everything the story's sources print, 10 sentences in 65 are finite claims. None of this bears on the theorems. What it offers their authors is small and concrete: publish the law file, and the one conditional step in an otherwise kernel-checked certificate becomes a step anyone can check.

Declaration of competing interest

The author declares no competing financial or personal interests. The authors of the paper, the formalisation and the post have not been consulted on this draft; the request for the law file is drafted and will be posted on the formalisation's repository.

CRediT authorship contribution statement

Carlos Toledo: conceptualization, methodology, software, validation, investigation, data curation, writing—original draft, writing—review and editing. The author list and the roles are to be agreed with the prospective co-authors.

Data availability

Every record, script and table source is public in the cert-machine repository (`https://github.com/carlostoledo1891/cert-machine`, commit 61aae8eb42f1; archived as Toledo, 2026a): the copies of the translator's records and pinned corpus `corpus/claimbox-riemann/` with their provenance `PROVENANCE-claimbox.json` (claimbox commit `b37d9800aa81428f7c3d16b8dfb97cdce80aef30`, 69 files, each with its sha256 on both ends); the manifest `LIFT-claimbox.json`; the battery `tools/check-claimbox-lift.js`; and the generator of this paper's numbers `tools/paper-numbers/riemann-finite-layer.js`. The translator, `claimbox` (Toledo, 2026b), holds the grammar, the readers, the runs and the recall at the commit cited. The sources are cited at their origin and pinned by digest in `corpus/claimbox-riemann/corpus/riemann/PINS.json`; the

`zeta-23-lean` repository's address redirected to `anthropics/formal-math` when checked on 2026-10-22, and is cited as pinned. The 11/12 manuscript is cited by digest and not redistributed.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used large language models (Claude, Anthropic) to draft code, exposition and this manuscript in a certified-verification setting: constructions and prose were proposed with the help of the models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for a record to be accepted. After using the tool the author reviewed and edited the content and takes full responsibility for the content of the publication. The records, not the models, are the evidence offered; they are public and re-runnable. For this paper specifically: the translator, its grammar and readers, the certificate files, the planted claims, the mathematician's claims with their derivations, the recall at scale and this prose were drafted by AI agents (Claude) under written briefs and consolidated into the public record; every verdict here is a computation by the checker over pinned bytes, and no reading by an agent enters any verdict. The two corrections of Section 3 are the machine correcting the agent.

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Appendix A. The battery

The sources. Twenty-five files were fetched byte for byte on 2026-10-22 and pinned by sha256 (Table A.3): `openai/math` at commit `1581d4cbdd69`, `anthropics/zeta-23-lean` at `e1a4e6508154`, the paper's e-print arXiv 2608.13637v2 (the LaTeX source, sha256 `ae369aaffb66`), the post as served (`82b95cd4d18c`), the video's captions as pasted (9,538 bytes, `b86fd8794cfb`). The sources were read for the claims they print and never run. Every decision took at most 0.014 seconds on an 8-core laptop.

The sentences read. Table A.4 lists the 10 of 65 sentences the grammar reads, and Table A.5 the count by author.

The planted claims. Fifteen claims about the certificate were written by Claude in the translator's session, as a claimant summarising the certificate might write them, some right and some wrong on purpose, each with the word expected of it recorded before the run (Table A.6). They are a planted battery, not a sample of what any published AI text says. 6 refuted, 4 certified, 5 refused with the rule, every one as expected. The one that measures

source	files	bytes
openai/math (the index, the scope note, the four Comparator statements, three READMEs, the history)	11	660,334
anthropics/zeta-23-lean (README, formalization.yaml, the audit, the Challenge files, the five PairCeiling files)	10	153,709
arXiv 2608.13637v2 (the e-print's members)	2	87,018
Anthropic's research post, as served	1	202,705
the video's captions, as pasted	1	9,538

Table A.3: The 25 pinned sources of the story, by origin.

sentence, as printed	author	file	shape
$2 - c_{\mathrm{MT}}^{-1} = 0.67250\ldots$	Claude: the paper, arXiv 2608.13637v2	arXiv/ paper-v5-draft18.tex	digits
$\frac{1}{2}(3 - c_{\mathrm{MT}}^{-1}) = 0.83625\ldots$	Claude: the paper, arXiv 2608.13637v2	arXiv/ paper-v5-draft18.tex	digits
$p_0 \le 0.6818287$	Claude: the paper, arXiv 2608.13637v2	arXiv/ paper-v5-draft18.tex	bound
$\lvert 256\,S(j) - j\rvert \le 3 \cdot 10^{-40}$	Claude: the paper, arXiv 2608.13637v2	arXiv/ paper-v5-draft18.tex	bound
$\lvert D(1)\rvert \le 0.82395317$	Claude: the paper, arXiv 2608.13637v2	arXiv/ paper-v5-draft18.tex	bound
$\lvert 256 \cdot S(j) - j\rvert \le 3 \cdot 10^{-40}$ $(0 < j < 256)$	Claude: the Lean formalization zeta23	zeta23/ README.md	bound
$2 - 1/c_1^* = 0.67250\ldots$	Claude: the Lean formalization zeta23	zeta23/ formalization.yaml	digits
$(3 - 1/c_1^*)/2 = 0.83625\ldots$	Claude: the Lean formalization zeta23	zeta23/ formalization.yaml	digits
$c_1^* = 0.7532960\ldots$	Claude: the Lean formalization zeta23	zeta23/ ChallengeDeps.lean	digits
$\lvert D(1)\rvert \le 82395317/10^8$	Claude: the Lean formalization zeta23	zeta23/PairCeiling/ LawN256.lean	bound

Table A.4: The 10 sentences the grammar reads.

author, source	sentences	read	refused
OpenAI: the manuscript's abstract, as the index prints it	6	0	6
OpenAI: the release (index, scope note, Lean statements, history)	7	0	7
Claude: the paper, arXiv 2608.13637v2	18	5	13
Claude: the Lean formalization zeta23	21	5	16
Anthropic: the research post	4	0	4
the video's captions	9	0	9

Table A.5: The 65 printed sentences by author.

a limit is the wrong last digit of the transcendental, $c_1^* = 0.7532961\ldots$: through the series certificate it is refuted; through the closed form it would be admitted and undecided.

claim	what it is	word
$\lvert D(1)\rvert \le 0.8239$	exact |D(1)| at the enclosures' ends is 0.82395316…, above 0.8239	REFUTED
$\lvert D(1)\rvert \le 0.824$	a rounding up: true	CERTIFIED
$\lvert D(1)\rvert < 0.82395316$	cut short by $7{\cdot}10^{-10}$	REFUTED
$\lvert 256 \cdot S(j) - j\rvert \le 2 \cdot 10^{-40}$ $(0 < j < 256)$	worst row is $1.84{\cdot}10^{-40}$	CERTIFIED
$\lvert 256 \cdot S(j) - j\rvert \le 1 \cdot 10^{-40}$ $(0 < j < 256)$	below the worst row	REFUTED
$\lvert 256 \cdot S(j) - j\rvert \le 3 \cdot 10^{-40}$ $(0 < j < 100)$	not the certificate's rows	REFUSED (count)
$\lvert 256 \cdot S(j) - j\rvert \le 3 \cdot 10^{-40}$ $(0 < j < 257)$	one row too many: the closed-band row is free	REFUSED (count)
$p_0 \le 0.68182$	$p_0 = 0.68182868\ldots$	REFUTED
$p_0 \le 0.6819$	a rounding up: true	CERTIFIED
$p_0 \ge 0.6818287$	relation the wrong way	REFUTED
$c_1^* = 0.7532961\ldots$	a wrong last digit of a transcendental: decided from the series certificate (mathematician.md §1), the enclosure 0.75329606785… lies below the box [0.75329605, 0.75329615] — the gap of the closed-form route closed by the reduction	REFUTED
$2 - 1/c_1^* = 0.6725\ldots$	true to four digits; decided from the series certificate	CERTIFIED
$D(1) \le 0.82395317$	a different quantity: no functional is written for D(1) without the bars	REFUSED (decider)
we checked that $\lvert D(1)\rvert \le 0.82395317$	a prefix: the grammar reads a claim at the head of its sentence	REFUSED (prose)
$1/(6 \cdot 256^2) + \tau/512 < 2.5431316 \cdot 10^{-6}$	no shape: an expression on the left of a relation is not a named quantity	REFUSED (prose)

Table A.6: The 15 planted claims and their words.

The mathematician's claims. The author of the translator's session then wrote 22 claims of its own, derived from the sources and the bytes, each with the word expected of it (Table A.7): the two eighth-digit bounds on c_1^*; the two "published as" prints; the CUE containment; the three pieces of the paper's ceiling arithmetic; the five ξ' identities, the four eight-digit and the four five-digit constants; and a control one unit above P_{flat}. 20 hold, 2 are refuted, both refutations expected: the 0.8362 print and the control.

Recall at scale. A defect planted in the bytes, the sentence or the page must come out as a refusal with its rule or as a changed decision, and an edit that is not a defect must leave the decision as it was. Every enclosure was moved by one unit either way and widened by one and two units; rows were dropped and exchanged; K and N were edited; the free row was moved; p_0 was edited in the file; the bar was moved in the sentence and in a copy of the page with the sentence following; another claimant's file was given as

claim	why the word was expected	word
$c_1^* \le 0.75329607$	the upper end of the enclosure, 0.753296067856070677 22…, is below the bar	CERTIFIED
$c_1^* \ge 0.75329606$	the lower end, 0.75329606785607067702…, is above the bar	CERTIFIED
$(3 - 1/c_1^*)/2$, published as 0.8362	the paper's abstract prints 0.8362; the exact value 0.836250351839705822… lies above the box [0.83615, 0.83625] its digits allow — a truncation (a rounding prints 0.8363); as a lower bound the print is still true	REFUTED
$2 - 1/c_1^*$, published as 0.6725	0.672500703… lies in [0.67245, 0.67255]	CERTIFIED
$\lvert 2 \cdot j \cdot K/256 - \mathrm{lo}_j - \mathrm{hi}_j \rvert \le 1$ $(0 < j < 256)$	every band enclosure contains the exact CUE value j·K/256 (124 at the upper end, 131 at the lower end)	CERTIFIED
$\lvert p_0 - 2/3 \rvert \le 0.016$	$p_0 - 2/3 = 0.0151620207\ldots$	CERTIFIED
$B(8.2) \le 0.6819$	$p_0 + 2.55{\cdot}10^{-6}{\cdot}8.2 = 0.68184954\ldots$	CERTIFIED
$c_{256} \le 2.55 \cdot 10^{-6}$	$1/(6{\cdot}256^2) + \tau/512 = 2.5431315104\ldots{\cdot}10^{-6}$	CERTIFIED
$J_{\mathrm{flat}} = 12517210259/88388425125$	the flat window integral, re-derived exactly from a_k and $\int$ s^d(1 − s)	CERTIFIED
$\kappa_9^{\mathrm{flat}} = 100905635384/88388425125$	1 + J_flat	CERTIFIED
$\varepsilon_9 = 1024/2990212875$	a_9·231/211 with a_9 = 2·4^10·9!/20!	CERTIFIED
$J_{\mathrm{quartic}} = 299652803309\,34234270211/285991090937\,677125000000$	the quartic window integral over my own autocorrelation coefficients; b_0 = $\int v^2$ and $\sum$ b_l = 0 beside it	CERTIFIED
$\kappa_9^{\mathrm{quartic}} = 277244140547\,469154168336/245053976636\,191319722125$	$(\int v^2$ + J_quartic$)/(\int v)^2$	CERTIFIED
$P_{\mathrm{flat}} \ge 0.85838371$	the Lean constant; margin $3.6{\cdot}10^{-9}$	CERTIFIED
$D_{\mathrm{flat}} \ge 0.92919185$	the Lean constant; margin $6.8{\cdot}10^{-9}$	CERTIFIED
$P_{\mathrm{quartic}} \ge 0.86864017$	the Lean constant	CERTIFIED
$D_{\mathrm{quartic}} \ge 0.93432008$	the Lean constant	CERTIFIED
$P_{\mathrm{flat}} \ge 0.85838$	the paper's print	CERTIFIED
$D_{\mathrm{flat}} \ge 0.92919$	the paper's print	CERTIFIED
$P_{\mathrm{quartic}} \ge 0.86864$	the paper's print	CERTIFIED
$D_{\mathrm{quartic}} \ge 0.93432$	the paper's print	CERTIFIED
$P_{\mathrm{flat}} \ge 0.85838372$	one unit of the eighth digit above the exact value 0.8583837135…	REFUTED

Table A.7: The mathematician's 22 claims, with the expectation recorded before each run.

the bytes; the last digit of c_1^* was changed in a copy of its file (Table A.8). The expectation of every row edit was computed from the edited integers by hand: an enclosure sits on one side of $j \cdot K/256$, which side varies by row, and τ is 1.63 units of $256/K$, so a move away from j crosses the tolerance and a move towards it does not. The first draft of the recall assumed every one-unit shift flips the decision; the first sample showed half do not, which is what the controls are for. 782 plants, 782 caught; 511 controls, 511 undisturbed; 1 undecided as recorded, the digits plant on the paper's closed-form claim (digits:af-mt-simple:0.67251). On the translator's earlier corpora the same test caught 56 of 56.

The translator's own battery, and its earlier corpora. The translator's test suite at the pinned commit prints 207 checks passing with 44 of 44 red controls firing, observed once from this repository on 2026-10-22 and kept

plant	must flip or refuse	must stay	undecided
one enclosure moved by ± 1 unit of K	255	255	0
one enclosure widened by one unit on one side	255	255	0
widened by two units	255	0	0
two enclosures exchanged	1	0	0
a row removed	2	0	0
the scale K edited	1	0	0
the count N edited	1	0	0
the free closed-band row moved by 10^{37}	1	1	0
the sentence edited: the bar moved, the range shortened, the relation flipped	5	0	0
the page copied with the bar moved and the sentence following	3	0	0
another claimant's file as the bytes	1	0	0
the exact rational p_0 edited in the file	1	0	0
the last printed digit of c_1^* changed in a copy of its file	1	0	1

Table A.8: The 1,294 rows of the recall at scale on the certificate.

with the copies. The stronger evidence for the pipeline is on its earlier corpora, in the method paper's register: EinsteinArena's 26-circle packing, decided as 6 predicates holding and one refuted on 3 pairs that overlap by 10^{-18}, float64 tangencies the platform admits and exact arithmetic does not; an earlier C_{1c} certificate of the optimization-constants registry, $C \geq 0.41030211724$, refuted by $3 \cdot 10^{-14}$, a lower bound rounded up; and a 605-point kissing record read two ways from one byte string, 9,510 against 9,480 violating pairs, the lesson that a reading is part of a verdict (Toledo, 2026c).

Appendix B. Reproduction

Each result re-derives with the command beside it in Table B.1: in the translator's repository at commit `b37d9800aa81`, with cert-machine beside it read-only, for the runs; in this repository for the copies and the numbers. Every run re-pins the sources before it reads them and refuses a file that does not match.

result	command (in `claimbox` unless marked)	expected
the pins	`make riemann && make drift`	25 files held
the certificate	`python3 -m claimbox batch corpus/ riemann-jobs.json --out records/ riemann --check --record records/ riemann-run.json`	Table 1
the planted claims	`python3 -m claimbox batch corpus/ riemann-ai-jobs.json --out records/ riemann-ai --check --record records/ riemann-ai-run.json`	Table A.6
the mathematician's claims	`python3 -m claimbox batch corpus/ riemann-math-jobs.json --out records/ riemann-math --check --record records/riemann-math-run.json`	Table A.7
recall at scale	`CLAIMBOX_FULL=1 python3 tests/riemann_recall.py`	782/782, 511/511
the translator's battery	`make test`	207 pass, 44/44 reds
the copies (cert-machine)	`node tools/check-claimbox-lift.js`	69 files held; the numbers identical
this paper's numbers (cert-machine)	`node tools/paper-numbers/ riemann-finite-layer.js`	the macro file rewritten identically

Table B.1: Each result, the command that re-derives it and what it must print.

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